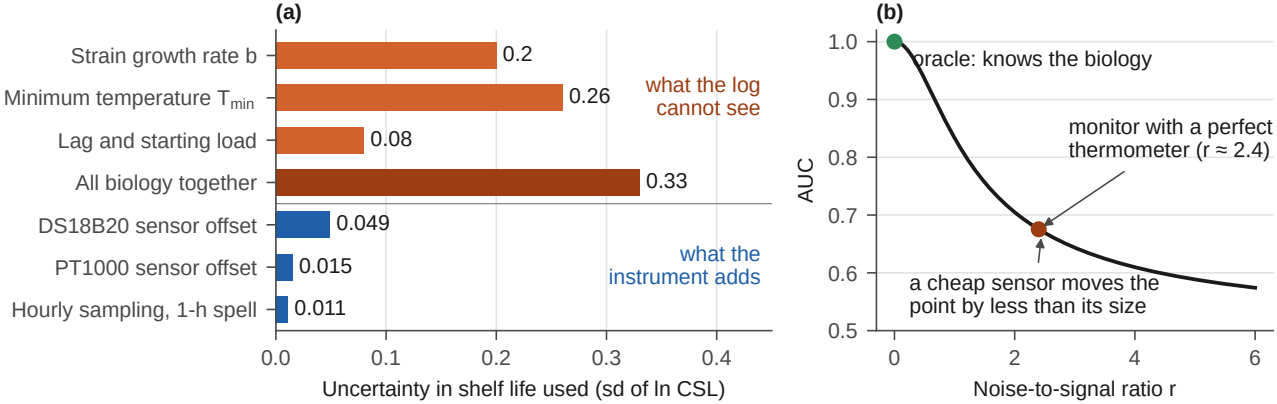
(a)
Strain growth rate b
Minimum temperature T_{min}
Lag and starting load
All biology together
DS18B20 sensor offset
PT1000 sensor offset
Hourly sampling, 1-h spell
0.2
0.26
0.08
0.33
0.049
0.015
0.011
what the log cannot see
what the instrument adds
0.0
0.1
0.2
0.3
0.4
Uncertainty in shelf life used (sd of ln CSL)
(b)
AUC
1.0
0.9
0.8
0.7
0.6
0.5
oracle: knows the biology
monitor with a perfect thermometer (r ≈ 2.4)
a cheap sensor moves the point by less than its size
0
2
4
6
Noise-to-signal ratio r